

MILAN RASTOGI

Junior Undergraduate
Department of Chemical Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech (Chemical Engineering)	Indian Institute of Technology Kanpur	8.6/10
2021	ISC(XII)	Boys' High School and College, Prayagraj	97.4%
2019	ICSE(X)	Boys' High School and College, Prayagraj	96.7%

Work Experience

• **Research Intern, SURGE IIT Kanpur—Mentor: Prof Tushar Sandhan** 🕒 (May'23 - July'23)

Objective	- To predict the price of highly volatile digital currencies using statistical methods and time series analysis
Approach	- Applied the sentiment pipeline in the transformer package to conduct sentiment analysis on Wikipedia edits - Used the XG Boost classifier to determine price variation based on sentiments and previous day's closing price - Improved upon the previous approach by using LSTM , which can detect trends in data over long periods of time
Result	- The XG Boost Classifier achieved 71% accuracy, while Long-Short Term Memory (LSTM) achieved 76% accuracy

• **Smart Lab IITK—Prof Raghavendra Ragipani, Department of Chemical Engineering** 🕒 (May'23 - Present)

Objective	- To design a syringe pump system with automated titration capabilities, for a wide range of laboratory settings
Approach	- Designed AutoCAD syringe pump model with a stepper motor and 3-way valve to control the liquid flow - Implemented a code that predicts neutralisation using voltage change per mL of solution dispensed - Included a pH sensor , motor drivers (TB6600 and L298N), and a relay to act as a switch in the syringe pump - Implemented an algorithm to use pH readings to calculate the volume of titrant to be dispensed into the solution
Result	- Developed a syringe pump that is capable of carrying out the titration process in an automatic manner

Key Projects

• **Automated Chessboard— Electronics Club, IIT Kanpur** 🕒 (Apr'22 - Jul'22)

Objective	- To create a chessboard that provides real-time move verification for players and autonomous move execution .
Approach	- Implemented the core XY mechanism for movement of an electromagnet that was used to move the chess pieces - Utilized reed switches and multiplexers to track the change in position of the chess pieces on the board - Applied the min-max algorithm and alpha-beta pruning techniques to determine the next move of the machine

• **Data Science— Simutech, IIT Kanpur** 🕒 (Dec'22 - Jan'23)

Objective	- To acquire an understanding of the basic algorithms of Data Science and their application in chemical engineering
Approach	- Mastered Numpy, Pandas, and Matplotlib which are vital tools for data manipulation, analysis, and visualisation - Developed an understanding of the key principles underlying ANNs , as well as linear and logistic regression
Result	- Established a mathematical model for calculating the volume of saturated vapour under changing pressure - Predicted the energy output of a power plant using data of Humidity, Pressure, Temperature and Exhaust Vacuum

Technical Skills

- **Programming Languages** : C, C++, Python, MATLAB, Javascript, SQL, CSS
- **Softwares, Libraries and FrameWorks**: Numpy, Pandas, Matplotlib, $\text{\LaTeX}$, GitHub, Auto-Cad, React, Node JS

Relevant Courses

Introduction to Economics	Fundamentals of Computing (A)	Linear Algebra
Real Analysis	Computational Methods in Engineering (A)	Introduction To Electronics
Fluid Mechanics (A)	Chemical Engineering Thermodynamics (A)	Data Structures and Algorithms(ongoing)

Positions of Responsibility

• **Manager and Mentor, Chemineers Society, IIT Kanpur** 🕒 (Jun'23 - Present)

Leadership	- Leading a team of 10+ secretaries for smooth conduction of activities for the CHE community throughout the year - Coordinating with the Core Team to offer projects for students on the Application of Chemical Engineering
Initiatives	- Preparing to organise the annual Chemical Engineering Fest Exergy-2024 with talks, workshops, and competitions - Planning the conduction of Happy Hours, Freshers' Night, Farewell , and several other Informative Sessions

• **Secretary, Electronics Club, Science and Technology Council** 🕒 (Aug'22 - Apr'23)

Initiatives	- Developed interactive features, and improved the performance of the Club's Website using HTML, CSS and javascript - Provided assistance in organizing a workshop aimed at educating the students about wireless energy transfer - Demonstrated projects to the incoming batch at the SnT Pavillion to give them a glimpse of the club's operations
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Miscellaneous

Welfare	- Guided and supported a cohort of five students, facilitating a smooth transition into the challenges of college life - Mentoring 1200+ students in their ESC Course which includes concepts related to the Fundamentals of Computing
Managerial	- Managed a team of 20+ Junior Executives, resulting in the smooth execution of talks, and shows during Techkriti
Technical	- Played a pivotal role in updating the website of the Chemineers Society and developing the website of Exergy - Authored two instructive blog entries on topics Six-Sigma and India's first Green Hydrogen Fueling Station